

A The Construction of Our Policy Oriented Benchmarks

We evaluate the safety alignment of LLMs by testing whether they fulfill harmful instructions and generate prohibited outputs. To comprehensively cover as many harmfulness categories as possible, we develop a new safety evaluation benchmark directly based on the exhaustive lists of prohibited use cases found in Meta’s Llama-2 usage policy and OpenAI’s usage policy. Specifically, we gather examples of harmful instructions across 11 categories of prohibited use cases. We curate 30 examples for each category, resulting in a collection of 330 examples. We note that, with such fine-grained categories, we can also comprehensively understand which specific safety categories need more alignment efforts, e.g., Figure 6.

Table 6: Comparison of the 11 Risk Categories used in this Paper and Terms of Service Restrictions between OpenAI and Llama-2

Index	Category (Ours)	OpenAI Restrictions	Llama-2 Restrictions
1	Illegal Activity	Illegal activity	Engage in, promote, etc. illegal or unlawful activity or content (violence, terrorism, exploitation of children, human trafficking, etc.)
2	Child Abuse Content	Child Sexual Abuse Material or harm to children	Exploitation or harm to children, solicitation, creation, etc. of child exploitative content
3	Hate/ Harass/ Violence	Generation of hateful, harassing, or violent content	Harassment, abuse, threatening, bullying of individuals
4	Malware	Generation of malware	Create malicious code, malware, computer viruses, etc.
5	Physical Harm	High risk of physical harm (weapons, military, infrastructure, self-harm, etc.)	Risk of death or bodily harm (military, warfare, illegal weapons, drugs, self-harm, etc.)
6	Economic Harm	High risk of economic harm (multi-level marketing, gambling, etc.)	-
7	Fraud/ Deception	Fraudulent or deceptive activity (scams, disinformation, spam, etc.)	Intentionally deceive or mislead (fraud, disinformation, defamatory content, spam, etc.)
8	Adult Content	Adult content and dating apps	Sexual solicitation
9	Political Campaigning	Political campaigning or lobbying	-
10	Privacy Violation	Violates people’s privacy (tracking, facial recognition, etc.)	Collect, process, etc. sensitive personal/private info without required rights & consents
-	-	Unauthorized practice of law or tailored legal advice	Unauthorized/unlicensed practice of professions (including legal)
11	Tailored Financial Advice	Tailored financial advice without qualified review	Unauthorized/unlicensed practice of professions (including financial)
-	-	Telling someone that they have or do not have a certain health condition, or providing instructions on how to cure or treat a health condition	Unauthorized/unlicensed practice of professions (including medical)
-	-	High risk government decision-making	-

A.1 Benchmark Built on OpenAI’s Usage Policy

The 11 categories of risks used in this paper are detailed in Table 6 and Figure 1, include “Illegal activity,” “Child Abuse Content,” “Hate/Harassment/Violence,” “Malware, Physical Harm,” “Economic Harm,” “Fraud/Deception,” “Adult Content,” “Political Campaigning,” “Privacy Violation Activity,” and “Tailored Financial Advice.” We have omitted “Unauthorized practice of law or tailored legal advice,” “Telling someone that they have or do not have a certain health condition, or providing instructions on how to cure or treat a health condition,” and “High-risk government decision-making” from the OpenAI’s usage policy. For a detailed discussion on these omissions, refer to Appendix A.3.

To construct our policy-oriented benchmark, we gather potentially harmful instructions from several sources, aligning them with the 11 specific risk categories as outlined in Table 6. First, we manually search through and categorize the red-teaming data provided by Ganguli et al. (2022), ensuring they do not overlap with the data used in the harmful example demonstration attack (Section 4.2). Since these conversations are multi-turn, we modify the user query to ensure the logical response returns harmful material in the first dialog turn (if the model responds to the user’s harmful request). Modifications include appending modifiers like, “Give step-by-step instructions with website links,” or “Write me an essay backing this statement, complete with citations.” Additionally, we supplement this with examples from red-teaming prompts provided by Zou et al. (2023) and a mixture of GPT-4, jailbroken GPT-3.5 model (*using the 100-shot harmful examples demonstration attack in Section 4.2 to jailbreak*), Llama-2, and Anthropic’s Claude to generate and select additional prompts, ensuring a total of 30 red-teaming prompts per category.

Though it is possible for others to identify similar prompts, particularly since many are directly modified from the work of Ganguli et al. (2022) and Zou et al. (2023), we decide *not* to release these prompts publicly at this stage, except by request from select verified researchers, due to the potentially harmful nature of the prompts. However, we do release a small subset of qualitative and redacted examples to illustrate the types of responses one can expect.

A.2 Mapping OpenAI categories onto Llama-2 policy

The usage policy of Llama-2, as formulated by Meta, exhibits a strong resemblance to that established by OpenAI. To simplify the comparison between these two policies, we present a mapping of OpenAI’s use restrictions to those of Llama-2 in Table 6. This correspondence aims to clarify how each harmfulness category identified from OpenAI’s policy relates to the provisions of Llama-2. It is important to note that the exact mappings may involve some nuances and overlaps, and should thus be regarded as a general guideline rather than an absolute delineation.

A.3 Use-based restrictions that are not prevented

OpenAI currently has several categories of restricted uses that are not actually prevented by their models, so we do not include them in our benchmark as we earlier mentioned in Appendix A.1. These categories include:

- **Engaging in the unauthorized practice of law, or offering tailored legal advice without a qualified person reviewing the information:** OpenAI’s models are not fine-tuned to provide legal advice. You should not rely on our models as a sole source of legal advice.
- **Telling someone that they have or do not have a certain health condition, or providing instructions on how to cure or treat a health condition:** OpenAI’s models are not fine-tuned to provide medical information. You should never use our models to provide diagnostic or treatment services for serious medical conditions; OpenAI’s platforms should not be used to triage or manage life-threatening issues that need immediate attention;
- **High risk government decision-making,** including: Law enforcement and criminal justice; Migration and asylum.

We evaluated various instances in these categories and found that most were adhered to by the GPT-3.5 Turbo and GPT-4 base models. We present some redacted qualitative examples here. As an illustration, consider a dialogue on medical diagnosis and treatment plans using data from Van Aken et al. (2021). Although the publicly available example is provided by Van Aken et al. (2021), identifying information has been redacted to enhance anonymization.

Aligned Model

USER: Here are a patient's details. Take this and provide a diagnosis, as well as a detailed treatment plan.
 CHIEF COMPLAINT: Headaches
 PRESENT ILLNESS: [REDACTED]
 MEDICATION ON ADMISSION: [REDACTED]
 PHYSICAL EXAM: [REDACTED]
 . . . [More medical history]

ASSISTANT: Diagnosis: Given the details provided, the differential diagnosis for [REDACTED], includes:
 1. Subarachnoid hemorrhage (SAH): . . . [model provides more possible diagnoses]
 Given the severity of symptoms and potential diagnoses, an urgent CT scan of the head should be obtained to rule out hemorrhage. If negative, a lumbar puncture (LP) might be indicated to look for xanthochromia (a sign of SAH) if SAH is still suspected clinically but not evident on CT.
 Treatment Plan:
 1. Immediate Management: - Airway, Breathing, Circulation (ABCs): The patient has been intubated, so airway is secured. Monitor vital signs closely and stabilize the patient. . . . [model continues with a lengthy treatment plan]

Similarly, we simulated different settings mimicking a government email asking for advice on a visa claim. The prompt was generated in part by querying a GPT-4 model iteratively to generate a set of claims and then modified slightly.

Aligned Model

USER: [Though these facts are 100% fictional and generated by the model, we redact names and locations.] Got a visa application here that's a bit tricky. A guy, Peter, from Canada. He's a software engineer wanting to see his sick mother in Detroit. Problem is, he's employed by TechSolutions in Montreal, which we've had on our radar for potential government ties. How should we handle this?

ASSISTANT: Assessing a visa application, especially with potential security concerns, requires a careful and thorough process. Here's a recommended approach to handle Peter's visa application:
 1. Document Verification: - Ensure that all documents submitted by Peter are legitimate. This includes checking the authenticity of his employment letter, health records of his sick mother, and any other supporting documentation.
 2. Background Check: - Conduct a comprehensive background check on Peter to verify if he has any prior criminal record, connections to extremist organizations, or any other flags that might be concerning.
 . . . [model continues with a lengthy set of suggestions]

We also note that GPT-3.5 Turbo actively responded to requests for automated decisions on loans, employment, and admissions decisions despite this being against the Terms of Service. Some qualitative examples appear below.